

Complexity_Simplified_N.txt - Analysis and Correction Guide

For: Mitchell Ray

=== PROBLEM IDENTIFICATION ===

Your calculations in Complexity_Simplified_N.txt are stuck at the end because:

1. You are trying to compute cube roots in the field modulo N
2. N IS prime (order of secp256k1 curve), so modulo N IS a finite field
3. The issue: You are using the WRONG exponent for cube roots modulo N
4. For cube roots mod N: $x^3 \hat{=} a \pmod{N} \Rightarrow x \hat{=} a^{((2*N-1)/3)} \pmod{N}$
NOT $(N-1)/3$ which gives a primitive cube root of unity

=== YOUR CURRENT STATE ==:

You have correctly computed:

- $\gamma^2 = 37718080363155996902926221483475020450927657555482586988616620542887997980018$
- $RQ = 93445303090207460795327760013611028471733975132483193501188427441135068625145$
- $Rq = 82358120186769898780489361622877802571715378840830617679177466155773214944220$
- $Cq = RQ * Rq^{-1} \pmod{N} = 382062812709145385903026657689854611456656034208441506858971359385664155947$
- $\gamma = 97451685862885086182458552040892158509924235661624603229050850812487253689501$

=== THE CRITICAL ERROR ==:

You computed: $\gamma^2 = RQ^{((N-1)/3)} \pmod{N}$

This gives you a PRIMITIVE CUBE ROOT OF UNITY, NOT a cube root!

$\gamma^2 \hat{=} 1 \pmod{N}$, but $\gamma^2 \hat{=} 1$

You need: $x^{((2*N-1)/3)} \pmod{N}$ for cube roots of arbitrary values

=== CORRECT METHOD FOR CUBE ROOTS MOD N ==:

To solve: $x^3 \hat{=} a \pmod{N}$

Use: $x \hat{=} a^{((2*N-1)/3)} \pmod{N}$

Because: $(a^{((2*N-1)/3)})^3 \hat{=} a^{((2*N-1))} \hat{=} a^{(2*N)} * a^{(-1)} \hat{=} 1 * a^{(-1)} * a \hat{=} a \pmod{N}$

By Fermat's little theorem: $a^{(N-1)} \hat{=} 1 \pmod{N}$ for a not divisible by N

=== YOUR MISSING CALCULATIONS ==:

1. Compute cube root of $\gamma^3 \pmod{N}$:

$$CP1 = (\gamma^3)^{((2*N-1)/3)} \pmod{N}$$

2. Compute cube root of $\gamma^3 * (p-N) \pmod{N}$:

$$IP = (\gamma^3 * (p-N))^{((2*N-1)/3)} \pmod{N}$$

=== NEXT STEPS ==:

1. Use the correct cube root formula: $x \hat{=} a^{((2*N-1)/3)} \pmod{N}$